

Shift Register Tile

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Module: shift_reg_tile | **Shuttle:** - | **Date:** 2026-04-20

Description

32-bit shift register with parallel load. Supports right shift by variable amount and synchronous parallel load.

Port Convention (SemiCoLab)

Port	Direction	Width	Description
clk	input	1	Clock
arst_n	input	1	Async reset, active low
csr_in	input	16	Control/Status Register input
data_reg_a	input	32	Operand A
data_reg_b	input	32	Operand B
data_reg_c	output	32	Result
csr_out	output	16	Control/Status Register output
csr_in_re	output	1	CSR input read enable
csr_out_we	output	1	CSR output write enable

Tile Port Usage

- csr_in[0] - shift enable
- csr_in[1] - load enable (takes priority over shift)
- data_reg_a - parallel load data (32-bit)
- data_reg_b - shift amount (bits [4:0], 0-31)
- data_reg_c - register output (32-bit)
- csr_out - mirrors csr_in[1:0] as status

Usage Guide

Assert `csr_in[1]` to load `data_reg_a` into the register. Assert `csr_in[0]` to shift right by `data_reg_b` positions.

Precheck Result

Stage	Result
Connectivity Check	PASS
Synthesis	PASS
Cell Count	236
Status	PASS

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